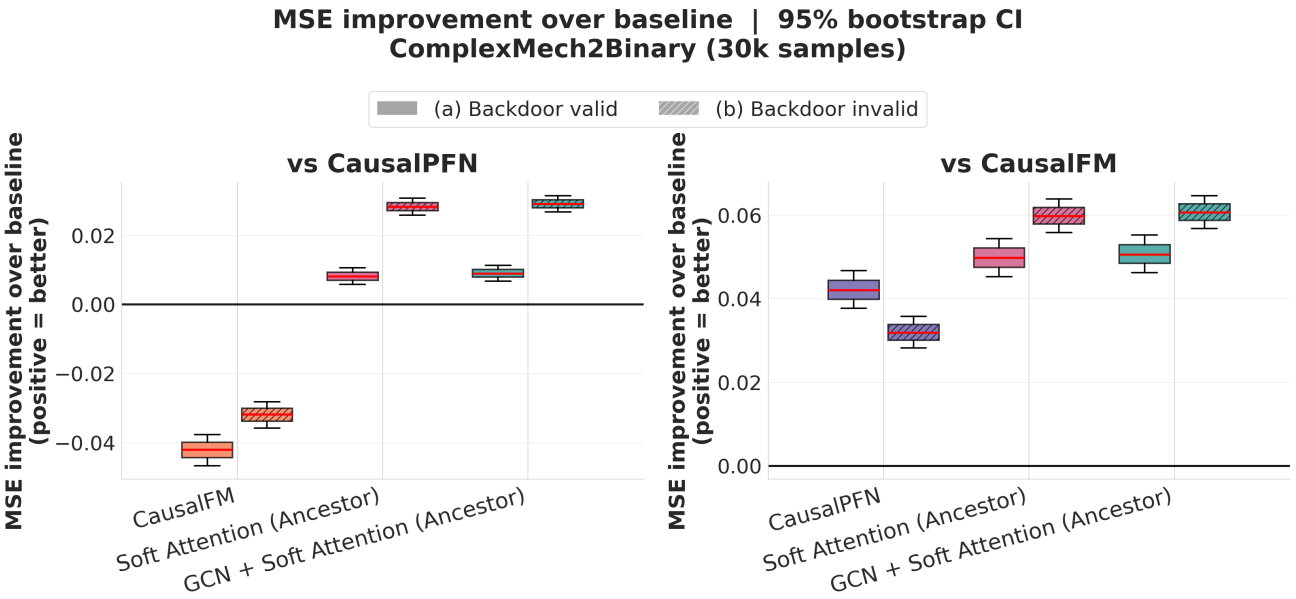

Figure 1. Performance under violation of the backdoor criterion on datasets sampled from our empirically validated complex-mechanisms prior. To enable comparison with CausalPFN and CausalFM, we use 30,000 datasets where the sampled SCM contains a strictly binary treatment variable. We consider two settings: (a) datasets where all observed covariates form a valid backdoor adjustment set for estimating the treatment effect (matching the assumptions for CausalFM and CausalPFN) and (b) datasets where this condition is violated. In both settings, our models receive only partial ancestral information, where the fraction of hidden entries in the partial ancestor matrix is sampled uniformly at random, matching the protocol used in our main experiments. We report the mean **difference** in performance compared to CausalPFN (left panel) and CausalFM (right panel). Errors are quantified via 95% bootstrap confidence intervals. **Left:** The Soft Attention (Ancestor) and GCN + Soft Attention (Ancestor) models trained on our prior perform slightly better than CausalPFN when the backdoor criterion is valid. When the backdoor criterion is invalid, the difference **increases substantially**, indicated by 0 on the y-axis (CausalPFN's performance) being far outside the confidence intervals; concretely, more than 22 standard errors. CausalFM is always significantly worse than CausalPFN. **Right:** The Soft Attention (Ancestor) and GCN + Soft Attention (Ancestor) models, as well as CausalPFN, significantly outperform CausalFM as a baseline. The difference for CausalPFN between cases (a) valid backdoor adjustment and (b) invalid backdoor adjustment is not statistically significant. The Soft Attention (Ancestor) and GCN + Soft Attention (Ancestor) models improve significantly over the baseline when the backdoor criterion is invalid.